MISH 44

SILICA EXPOSURE CONTROL

AGC CSEA Alignment

Risk Identification and Analysis

OSHA Core Element

Hazard Prevention and Control

Section Summary

Defines silica exposure control measures including exposure assessment, engineering controls, respiratory protection, and medical surveillance per OSHA Table 1.

Core Values: Integrity | Honesty | Professionalism | Thoroughness

"Building projects for the client, NOT the dollar."

SILICA EXPOSURE CONTROL

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AGC CSEA Alignment	OSHA Core Element
Risk Identification and Analysis	Hazard Prevention and Control

1.0 PURPOSE

To establish the requirements, responsibilities, and procedures for the proper handling of respirable crystalline silica. Further, to ensure compliance with the OSHA/Washington Industrial Safety and Health Act (WISHA) standards.

2.0 RESPONSIBILITY

The provisions of this procedure apply to all of MH Construction's personnel and lower-tier contractors performing work efforts at construction work locations where exposure to crystalline silica is possible.

2.1 MH Construction Safety Department shall be responsible for the following:

2.1.1 Providing technical consultation and advice regarding the handling and exposure to respirable crystalline silica.

2.2 The Key Supervisor shall be responsible for implementing and ensuring compliance with this procedure, developing and implementing exposure assessment to respirable silica.

2.3 Employees shall be responsible for complying with the provisions of this procedure.

3.0 DEFINITIONS

Action level. A concentration of airborne respirable crystalline silica of 25 µg/m3, calculated as an 8-hour time-weighted average (TWA).

Competent person. An individual who is capable of identifying existing and foreseeable respirable crystalline silica hazards in the workplace and who has authorization to take prompt corrective measures to eliminate or minimize them. The competent person must have the knowledge and ability necessary to fulfill the responsibilities set forth in WAC 296-840-140.

Construction work. All or any part of excavation, construction, erection, alteration, repair, demolition, and dismantling, of buildings and other structures and all operations in connection therewith; the excavation, construction, alteration and repair of sewers, trenches, caissons, conduits, pipe lines, roads and all operations pertaining thereto; the moving of buildings and other structures, and to the construction, alteration, repair, or removal of wharfs, docks, bridges, culverts, trestles, piers, abutments or any other construction, alteration, repair or removal work related thereto.

Director. The director of the department of labor and industries or his/her authorized representative.

DOSH. The division of occupational safety and health, Washington state department of labor and industries.

Employee exposure. The exposure to airborne respirable crystalline silica that would occur if the employee were not using a respirator.

High-efficiency particulate air (HEPA) filter. A filter that is at least 99.97 percent efficient in removing mono-dispersed particles of 0.3 micrometers in diameter.

Objective data. Information, such as air monitoring data from industry-wide surveys or calculations based on the composition of a substance, demonstrating employee exposure to respirable crystalline silica associated with a particular product or material or a specific process, task, or activity. The data must reflect workplace conditions closely resembling or with a higher exposure potential than the processes, types of material, control methods, work practices, and environmental conditions in the employer's current operations.

Permissible exposure limit (PEL). A concentration of airborne respirable crystalline silica of 50 µg/m3, calculated as an 8-hour TWA.

Physician or other licensed health care professional (PLHCP). An individual whose legally permitted scope of practice (i.e., license, registration, or certification) allows him or her to independently provide or be delegated the responsibility to provide some or all of the particular health care services required by WAC 296-840-145.

Regulated area. An area, demarcated by the employer, where an employee's exposure to airborne concentrations of respirable crystalline silica exceeds, or can reasonably be expected to exceed, the PEL.

Respirable crystalline silica. Quartz, cristobalite, and/or tridymite contained in airborne particles that are determined to be respirable by a sampling device designed to meet the characteristics for respirable-particle-size-selective samplers specified in the International Organization for Standardization (ISO) 7708:1995: Air quality – Particle size fraction definitions for health-related sampling.

Specialist. An American Board Certified Specialist in Pulmonary Disease or an American Board Certified Specialist in Occupational Medicine.

4.0 EXPOSURE ASSESSMENT

This section applies to all occupational exposures except for tasks performed according to the requirements in WAC 296-840-110.

(1) Permissible exposure limit (PEL). You must ensure that no employee is exposed to an airborne concentration of respirable crystalline silica in excess of the 8-hour TWA. (2) Exposure assessment. You must assess the exposure of each employee who is or may reasonably be expected to be exposed to respirable crystalline silica at or above the action level in accordance with either the performance option in (a) of this subsection or the scheduled monitoring option in (b) of this subsection.

(a) Performance option. You must assess the 8-hour TWA for each employee on the basis of any combination of air monitoring data or objective data sufficient to accurately characterize employee exposures to respirable crystalline silica. (b) Scheduled monitoring option:

(i) You must perform initial monitoring to assess the 8-hour TWA for each employee on the basis of personal breathing zone air samples that reflect the exposures of employees on each shift, for each job classification, in each work area. (ii) Where several employees perform the same tasks on the same shift and in the same work area, you may sample a representative fraction of these employees in order to meet this requirement.

(iii) In representative sampling, you must sample the employee(s) who are expected to have the highest exposure to respirable crystalline silica. (iv) If initial monitoring indicates that employee exposures are below the action level, you may discontinue monitoring for those employees whose exposures are represented by such monitoring. (v) Where the most recent exposure monitoring indicates that employee exposures are at or above the action level but at or below the PEL, you must repeat such monitoring within six months of the most recent monitoring. (vi) Where the most recent exposure monitoring indicates that employee exposures are above the PEL, the employer shall repeat such monitoring within three months of the most recent monitoring. (vii) Where the most recent (noninitial) exposure monitoring indicates that employee exposures are below the action level, you must repeat such monitoring within six months of the most recent monitoring until two consecutive measurements, taken seven or more days apart, are below the action level, at which time the employer may discontinue monitoring for those employees whose exposures are represented by such monitoring, except as otherwise provided in subsection (3) of this section. (3) Reassessment of exposures. You must reassess exposures whenever a change in the production, process, control equipment, personnel, or work practices may reasonably be expected to result in new

or additional exposures at or above the action level, or when the employer has any reason to believe that new or additional exposures at or above the action level have occurred. (4) Methods of sample analysis. You must ensure that all samples taken to satisfy the monitoring requirements of subsection (2) of this section are evaluated by a laboratory that analyzes air samples for respirable crystalline silica in accordance with the procedures in WAC 296-840-165 Appendix A—Methods of sample analysis of this chapter. (5) Employee notification of assessment results.

(a) Individually notify each affected employee in writing of the results of that assessment or post the results in an appropriate location accessible to all affected employees in accordance with the following:

(i) Construction employers must notify affected employees within five working days after receiving any results of an exposure assessment in accordance with this rule. (ii) All other employers must notify affected employees within fifteen working days after receiving any results of an exposure assessment in accordance with this rule.

(b) Whenever an exposure assessment indicates that employee exposure is above the PEL, you must describe in the written notification the corrective action being taken to reduce employee exposure to or below the PEL.

(6) Observation of monitoring.

(a) Where air monitoring is performed to comply with the requirements of this chapter, you must provide affected employees or their designated representatives an opportunity to observe any monitoring of employee exposure to respirable crystalline silica. (b) When observation of monitoring requires entry into an area where the use of protective clothing or equipment is required for any workplace hazard, you must provide the observer with protective clothing and equipment at no cost and shall ensure that the observer uses such clothing and equipment.

[Statutory Authority: Chapter RCW 49.17.010, .040, .050, and .060. 18-07-098 (Order 16-20), § 296-840-105, filed 03/20/2018, effective 04/23/2018.]

Table 1 Specified Exposure Control Methods When Working With Materials

Table 1 Specified Exposure Control Methods When Working With Materials Containing Crystalline Silica

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Table 1 Specified Exposure Control Methods When Working With Materials Containing Crystalline Silica Equipment/ Task Engineering and Work Practice Control Methods Required Respiratory Protection and Minimum Assigned Protection Factor (APF) (xviii) Heavy equipment and utility vehicles for tasks such as grading and excavating but not including: Demolishing, abrading or fracturing silica-containing materials Apply water and/or dust suppressants as necessary to minimize dust emissions. OR When the equipment operator is the only employee engaged in the task, operate equipment from within an enclosed cab. None

None None

None Statutory Authority: Chapter RCW 49.17.010, .040, .050, and .060. 18-07-098 (Order 16-20), § 296-840-110, filed 03/20/2018, /23/2018.]

5.0 Respiratory Protection

5.1 Where respiratory protection is required by this section, you must provide each employee an appropriate respirator that complies with the requirements of this section and chapter 296-842 WAC, Respirators.

- Respiratory protection is required:

■ Where exposures exceed the PEL during periods necessary to install or implement feasible engineering and work practice controls; ■ Where exposures exceed the PEL during tasks, such as certain maintenance and repair tasks, for which engineering and work practice controls are not feasible; ■ During tasks for which an employer has implemented all feasible engineering and work practice controls and such controls are not sufficient to reduce exposures to or below the PEL; ■ During periods when an employee or employees are in a regulated area; and ■ Where specified for tasks performed according to the requirements in Table 1 of WAC 296-840-110 Specified exposure control methods.

6.0 Respiratory protection program.

(1) Where respirator use is required by this chapter, you must institute a respiratory protection program in accordance with chapter 296-842 WAC, Respirators.

(2)Specified exposure control methods. For tasks listed in Table 1 of WAC 296-840-110, if you fully and properly implement the engineering controls, work practices, and respiratory protection described in Table 1 of WAC 296-840-110, you will be considered to be in compliance with WAC 296-840-125(1) and the requirements for selection of respirators in chapter 296-842 WAC with regard to exposure to respirable crystalline silica.

- 7.0 Written exposure control plan
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- You must establish and implement a written exposure control plan that contains at least the following elements:

■ A description of the tasks in the workplace that involve exposure to respirable crystalline silica; ■ A description of the engineering controls, work practices, and respiratory protection used to limit employee exposure to respirable crystalline silica for each task; and ■ A description of the housekeeping measures used to limit employee exposure to respirable crystalline silica.

- You must review and evaluate the effectiveness of the written exposure control plan at least annually and update it as necessary.
- You must make the written exposure control plan readily available for examination and copying, upon request, to each employee covered by this chapter, their designated representatives, and the director.
- In addition to the above written exposure control plan requirements, you must include the following elements for construction work:
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(a) A description of the procedures used to restrict access to work areas, when necessary, to minimize the number of employees exposed to respirable crystalline silica and their level of exposure, including exposures generated by other employers or sole proprietors. (b) A competent person to make frequent and regular inspections of job sites, materials, and equipment to implement the written exposure control plan.

8.0 Recordkeeping.

- Air monitoring data.

■ You must make and maintain an accurate record of all exposure measurements taken to assess employee exposure to respirable crystalline silica, as prescribed in WAC 296-840-105(2). This record must include at least the following information: ■ The date of measurement for each sample taken; ■ The task monitored; ■ Sampling and analytical methods used; ■ Number, duration, and results of samples taken; ■ Identity of the laboratory that performed the analysis; ■ Type of personal protective equipment, such as respirators, worn by the employees monitored; and

(vii) Name, Social Security number, and job classification of all employees represented by the monitoring, indicating which employees were actually monitored. (b) You must ensure that exposure records are maintained and made available in accordance with chapter 296-802 WAC, Employee medical and exposure records.

- Objective data.

■ You must make and maintain an accurate record of all objective data relied upon to comply with the requirements of this chapter. This record must include at least the following information: ■ The crystalline silica-containing material in question; ■ The source of the objective data; ■ The testing protocol and results of testing; ■ A description of the process, task, or activity on which the objective data were based; and ■ Other data relevant to the process, task, activity, material, or exposures on which the objective data were based. ■ You must ensure that objective data are maintained and made available in accordance with chapter 296-802 WAC, Employee medical and exposure records.

- Medical surveillance.

■ You must make and maintain an accurate record for each employee covered by medical surveillance under WAC 296-840-145. The record must include the following information about the employee: ■ Name and Social

Security number; ■ A copy of the PLHCPs' and specialists' written medical opinions; and ■ A copy of the information provided to the PLHCPs and specialists. ■ You must ensure that medical records are maintained and made available in accordance with chapter 296-802 WAC, Employee medical and exposure records. [

WRITTEN MEDICAL REPORT FOR EMPLOYEE

EMPLOYEE NAME:_____

_____

DATE OF EXAMINATION:

_____

TYPE OF EXAMINATION:_____

_____

[ ] Initial examination [ ] Periodic examination [ ] Specialist examination [ ] Other:

RESULTS OF MEDICAL EXAMINATION:

Physical Examination – [ ] Normal [ ] Abnormal (see below) [ ] Not performed Chest X-Ray – [ ] Normal [ ] Abnormal (see below) [ ] Not performed Breathing Test (Spirometry) –[ ] Normal [ ] Abnormal (see below) [ ] Not performed Test for Tuberculosis – [ ] Normal [ ] Abnormal (see below) [ ] Not performed

Other: [ ] Normal [ ] Abnormal (see below) [ ] Not performed

Results reported as abnormal:

[ ] Your health may be at increased risk from exposure to respirable crystalline silica due to the following:

RECOMMENDATIONS:

[ ] No limitations on respirator use [ ] Recommended limitations on use of respirator: _ [ ] Recommended limitations on exposure to respirable crystalline silica: _ _

Dates for recommended limitations, if applicable: to_____

MM/DD/YYYY MM/DD/YYYY

[ ] I recommend that you be examined by a Board Certified Specialist in Pulmonary Disease or Occupational Medicine [ ] Other recommendations*: Your next periodic examination for silica exposure should be in: [ ] 3 years [ ] Other:_____

MM/DD/YYYY

Examining Provider:_____ (signature) Provider Name: Office Address:_____ Date: Office Phone:_____

WRITTEN MEDICAL OPINION FOR EMPLOYER

EMPLOYER:

EMPLOYEE NAME:_ DATE OF EXAMINATION: TYPE OF EXAMINATION:

[ ] Initial examination[ ] Periodic examination [ ] Specialist examination [ ] Other:

USE OF RESPIRATOR:

[ ] No limitations on respirator use [ ] Recommended limitations on use of respirator: _ [ ] Recommended limitations on exposure to respirable crystalline silica: _ _

Dates for recommended limitations, if applicable: to_____

MM/DD/YYYY MM/DD/YYYY

The employee has provided written authorization for disclosure of the following to the employer (if applicable): [ ] This employee should be examined by an American Board Certified Specialist in Pulmonary Disease or Occupational Medicine [ ] Recommended limitations on exposure to respirable crystalline silica:_

Dates for exposure limitations noted above: to_____

MM/DD/YYYY MM/DD/YYYY

NEXT PERIODIC EVALUATION: [ ] 3 years [ ] Other: _____

MM/DD/YYYY

Examining Provider:_____ (signature)
Provider Name: Office Address:

Date: _____

Provider's specialty: Office Phone:

[ ] I attest that the results have been explained to the employee. The following is required to be checked by the Physician or other Licensed Health Care Professional (PLHCP): [ ] I attest that this medical examination has met the requirements of the medical surveillance

MH CONSTRUCTION SILICA WORK PLAN

SIGN-OFF LOG

NAME (print) SIGNATURE SOCIAL SECURITY # DATE _____

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